Spartan UltraScale+ FPGAs SelectIO Resources User Guide (UG861)

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ISERDESE3

The ISERDESE3 element is available to perform input deserialization for designs migrating from previous FPGA families or for designs not requiring native mode primitives. The ISERDESE3 in Spartan UltraScale+ devices is a serial-to-parallel converter with specific clocking and logic features designed to facilitate the implementation of high-speed source-synchronous applications. The ISERDESE3 avoids the additional timing complexities encountered when designing deserializers in the device logic.
There are some differences between the ISERDESE3 and its predecessors. ISERDESE3 does not have:

- BITSLIP input performing a bitslip operation synchronous to CLKDIV.
- Selectable CE inputs able to function as a 2:1 serial-to-parallel converter clocked by CLKDIV.
- OFB input, being a direct connection between the OSERDES serial output and this input.
- SHIFTIN and SHIFTOUT pins allow extending the deserialization capability up to 14 bits by cascading two ISERDES using direct connections.

The ISERDESE3 can deserialize an incoming signal by 2 or 4 in SDR data capture, and by 4 or 8 in DDR data capture mode. When used for SDR data capture, the valid outputs are every other data output pin. For example, when used as a 1:4 deserializer using an SDR clock, the data width should be set to 8, and the data received is taken from Q0, Q2, Q4, and Q6. Details of which SerDes output pins to use and which value to apply to the DATA_WIDTH attribute are shown in the following table.

★ **Tip:** The first serial bit received in a word is Q0.

Table: ISERDESE3 Output Connections in SDR and DDR Modes

SDR or DDR	Required Ratio	DATA_WIDTH Attribute to Apply	Data Output Pins to Use
DDR	1:8	8	Q7, Q6, Q5, Q4, Q3, Q2, Q1, Q0
DDR	1:4	4	Q3, Q2, Q1, Q0
SDR	1:8	N/A	N/A
SDR	1:4	8	Q6, Q4, Q2, Q0
SDR	1:2	4	Q2, Q0

Other deserialization ratios and word alignment schemes are possible through the use of additional logic resources in the FPGA logic (see *Bitslip in Logic* (XAPP1208). The ISERDESE3 also contains a shallow eight entry FIFO that can optionally be used for clock domain transfers. When not used, the FIFO control signals should be connected to GND. When using the FIFO, the FIFO_RD_EN should be driven by the inverted FIFO_EMPTY signal to ensure the FIFO_write and read pointers do not overlap every eight clock cycles. As shown in the following figures, latency through the FIFO depends on FIFO_RD_CLK. When the write pointer is updated early with respect to FIFO_RD_CLK the latency through the FIFO is shorter.
Because clock routing can vary, the MMCM with the ZHOLD compensation compensates for the clock routing. To ensure all of the clock outputs from the MMCM are properly compensated for, the CLOCK_DELAY_GROUP must be used (see Figure 2). To ensure the ISERDES is properly aligned after a reset, see Component Mode Reset Sequence. When the clocks are not compensated for, such as when clock-capable inputs are directly connected to clock buffers (BUFG, BUFGCE, BUFGCE_DIV), additional bitslip logic is required. See *Bitslip in Logic* (XAPP1208).

Figure: ISERDES FIFO Latency with DATA_WIDTH = 8

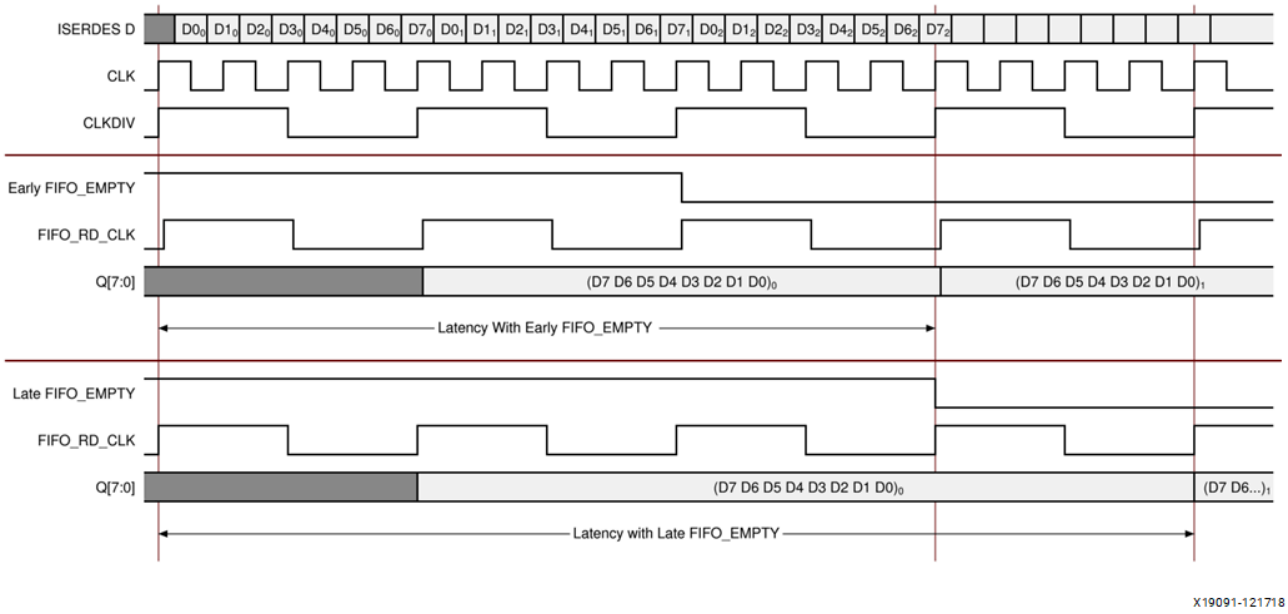
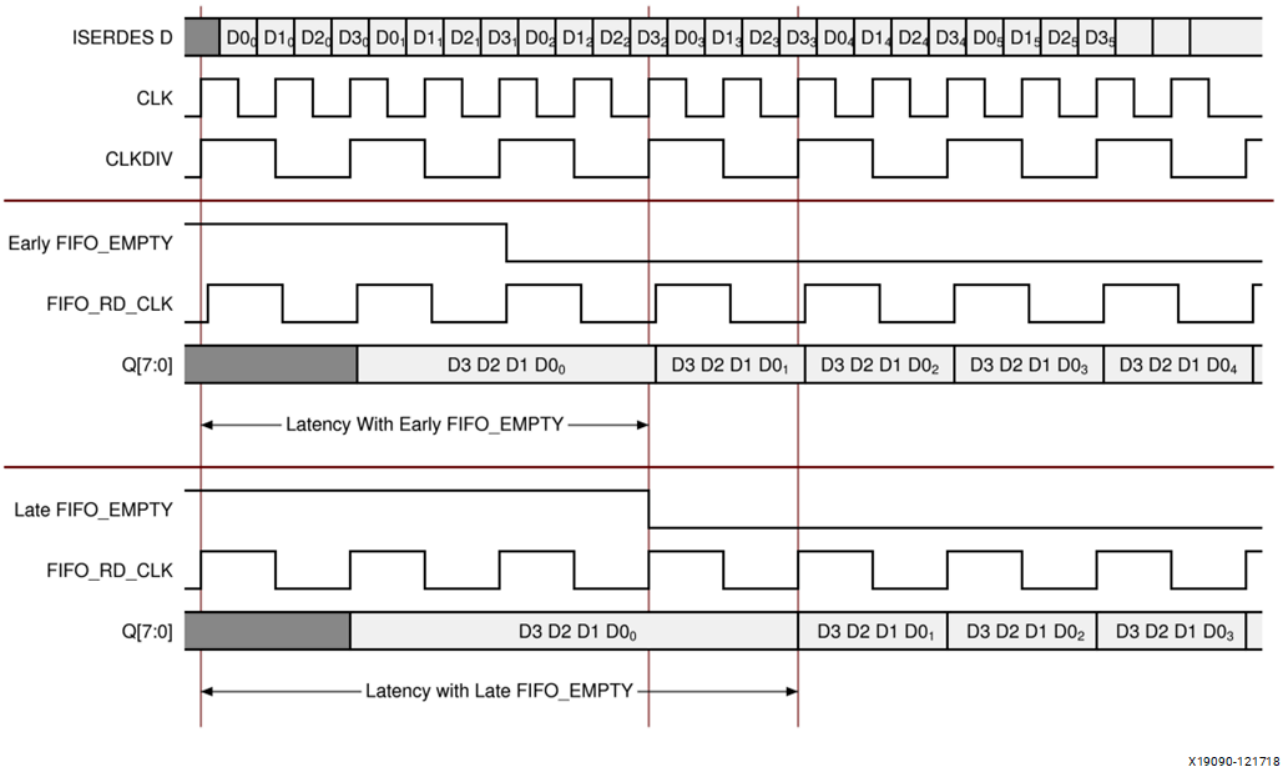


Figure: ISERDES FIFO Latency with DATA_WIDTH = 4



ISERDESE3 Ports

The following table lists the ISERDESE3 ports.

Table: ISERDESE3 Ports

Port	I/O	Type	Description
CLK	Input	Clock	High-speed clock input. Clock serial input data stream.
CLK_B	Input	Clock	Inverted version of CLK when IS_CLK_INVERTED=0 and IS_CLK_B_INVERTED=0.
CLKDIV	Input	Clock	Low-speed divided clock input.
D	Input	Data	Serial input data Synchronous to CLK/CLK_B.
Q[7:0]	Output	Data	Registered outputs. Synchronous to FIFO_RD_CLK when FIFO_ENABLE is TRUE.
RST	Input	Reset	Asynchronous reset.

Port	I/O	Type	Description
FIFO_RD_CLK	Input	Clock	FIFO read clock.
FIFO_RD_EN	Input	Enable	Enables reading the FIFO when asserted.
FIFO_EMPTY	Output		Indicates the FIFO is empty when asserted.
INTERNAL_DIVCLK	Output	Clock	Reserved

ISERDESE3 Attributes

The following table lists the ISERDESE3 attributes.

Table: ISERDESE3 Attributes

Attribute	Values	Default	Type	Description
DATA_WIDTH	4 or 8	8	Decimal	Defines the serial-to-parallel converter width.
FIFO_ENABLE	TRUE/FALSE	FALSE	String	The FIFO is used when the attribute is set TRUE and bypassed when the attribute is set FALSE.
FIFO_SYNC_MODE	TRUE/FALSE	FALSE	String	Set to FALSE when the ISERDES internal FIFO write clock and the FIFO read clock accessed from FPGA logic are from separate or common clock domains. This is the preferred selection because it supports all clocking options. TRUE: Reserved for later use.
IS_CLK_INVERTED	1 or 0	0	Bit	Sets a local clock inversion for CLK input.
IS_CLK_B_INVERTED	1 or 0	0	Bit	Sets a local clock inversion for CLK_B input. When IS_CLK_B_INVERTED=1, CLK

Attribute	Values	Default	Type	Description
				and CLK_B must be driven by the same global clock buffer. When IS_CLK_B_INVERTED=0, CLK_B must be driven by the same global clock buffer as CLK through an inverter.
IS_RST_INVERTED	1 or 0	0	Bit	Sets a local inversion for RST input when 1.
SIM_DEVICE	ULTRASCALE, ULTRASCALE_PLUS	ULTRASCALE	String	Device family for behavioral simulation.

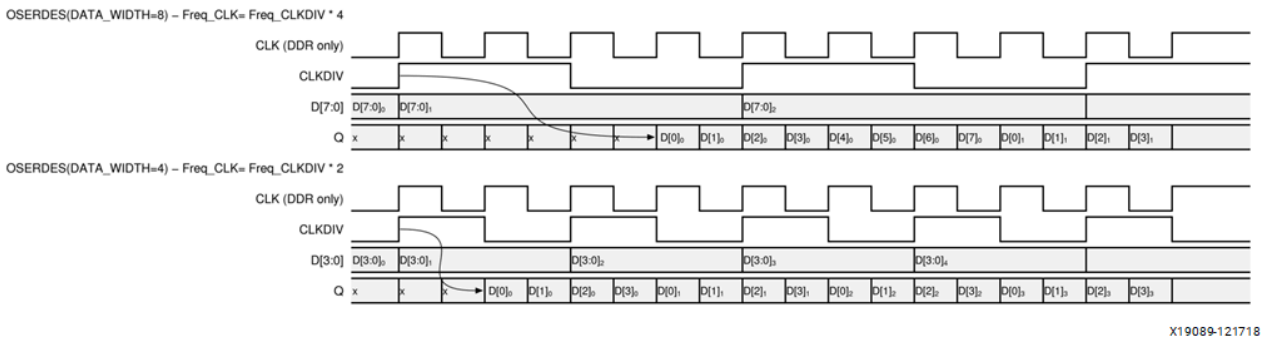
OSERDESE3

The OSERDESE3 primitive is available to perform output serialization for designs migrating from previous FPGA families or for designs not requiring native mode primitives. The OSERDESE3 in Spartan UltraScale+ devices is a 4- or 8-bit parallel-to-serial converter with specific clocking features to facilitate the implementation of source-synchronous and other applications. If other serial-to-parallel conversion factors are required, use the ODDRE1 primitive or implement a gearbox in internal logic.
There are some differences between the OSERDESE3 and its predecessors. The following functionality is not available in the OSERDESE3:

- OCE input enable pin for the serial output of the OSERDES.
- SHIFTIN and SHIFTOUT pins can use local dedicated connections to extend the serialization capabilities of the OSERDES.
- OFB output providing a straight and direct connection between the OSERDES output and ISERDES input without using an input and/or output buffer (IOB) and pin.
- Parallel 3-state and serial TBYTE functionality.

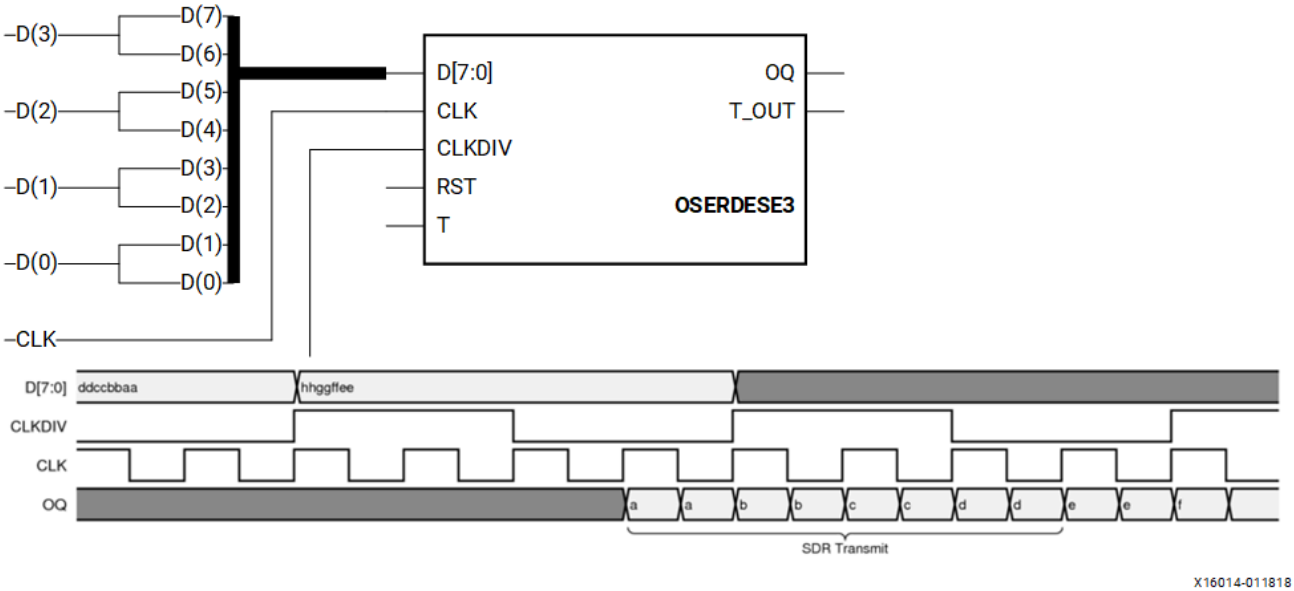
The latency through the OSERDES depends on the DATA_WIDTH setting as shown in the following figure.

Figure: OSERDES Latency



The OSERDESE3 can serialize an outgoing signal by a 2 or 4 in SDR mode, or by a 4 or 8 in DDR mode. When used with SDR clocking, the DATA_WIDTH attribute is to be set to twice the desired width and data to be transmitted should be applied to two pins at a time. See the following figure.

Figure: OSERDES Used in x4 SDR Mode (DATA_WIDTH = 8)



The full range of possibilities together with the associated attribute settings and required connections are shown in the following table.

★ **Tip:** The data applied to SerDes input D0 is the first bit to be transmitted in all cases.

Table: OSERDESE3 Output Connections in SDR and DDR Modes

SDR or DDR	Required Ratio	DATA_WIDTH Attribute Applied to OSERDESE3	Data Bits to be Applied to OSERDESE3
DDR	8:1	8	D7, D6, D5, D4, D3, D2, D1, D0
DDR	4:1	4	0, 0, 0, 0, D3, D2, D1, D0
SDR	8:1	8	N/A
SDR	4:1	4	D3, D3, D2, D2, D1, D1, D0, D0

SDR or DDR	Required Ratio	DATA_WIDTH Attribute Applied to OSERDESE3
SDR	2:1	0, 0, 0, 0, D1, D1, D0, D0

OSERDESE3 Ports

The following table lists the OSERDESE3 ports.

Table: OSERDESE3 Ports

Port	I/O	Descrip
CLK	Input	High-speed clock input
CLKDIV	Input	Low-speed divided clock input
D[7:0]	Input	Parallel data inputs for serializa synchron to CLKDIV
OQ	Output	Datapatl output
RST	Input	Asynchr reset.
T_OUT	Output	3-state control output

Port	I/O	Descrip
		to IOB
T	Input	3-state input from internal logic, combina 3-state T to T_OUT path. A logic High means the data is 3-stated and a logic Low means the data is not 3-stated.

OSERDESE3 Attributes

The following table lists the OSERDESE3 attributes.

Table: OSERDESE3 Attributes

Attribute	Values	Default	Type	Description
DATA_WIDTH	4 or 8	8	Decimal	Defines the parallel-to-serial data converter width.
INIT	1 or 0	0	Binary	Initializes the OSERDESE3 flip-flops to the value specified.
ODDR_MODE	TRUE/FALSE	FALSE	String	Internal property set by the Vivado tools when using ODDRE1. Do not modify. Forces the OSERDESE3 into an ODDRE1 mode with a 3-state ODDRE1 flip-flop as shown in ODDR with Serialized 3-State. In the ODDRE1 mode, data is connected to D[4,0] and the clock must be connected to CLK. The ODDRE1 primitive is recommended.
OSERDES_D_BYPASS	TRUE/FALSE	FALSE	String	When TRUE, D[0] is passed onto OQ. When FALSE, serialized D[0] and D[4] is output on OQ.
OSERDES_T_BYPASS	TRUE/FALSE	FALSE	String	When TRUE, D[1] is passed onto T_OUT. When FALSE, serialized D[1] and D[5] is output on T_OUT.
IS_CLK_INVERTED	1 or 0	0	Bit	Sets a local clock inversion for CLK input when 1.

Attribute	Values	Default	Type	Description
IS_CLKDIV_INVERTED	1 or 0	0	Bit	Sets a local clock inversion for CLKDIV input when 1.
IS_RST_INVERTED	1 or 0	0	Bit	Sets a local inversion for RST input when 1.
SIM_DEVICE	ULTRASCALE_PLUS	ULTRASCALE	String	Device family for behavioral simulation.